

Ariva Scale → TARE Tablet

The short version

1. Power the scale from its own **12 V wall adapter** (it is not USB-powered).
2. Run the scale's **RJ50→USB cable (#72256236)** into a **USB-C OTG hub**, then into the tablet's USB-C port.
3. On the scale, set **Comm Type = USB Virtual COM** and **Protocol = 8217 Mettler-Toledo**.
4. In the app, **Settings** → **Weight source** → **Serial (Streaming)** and match the serial values.
5. Place a weight — it flows into the shared scale reading and every order checks live.

BILL OF MATERIALS

What you need to buy

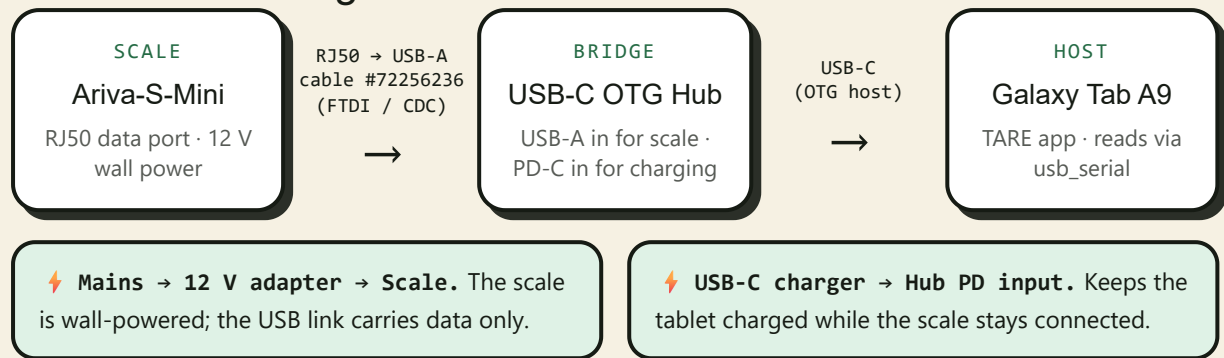
One-station shopping list. Quantities are per pack station.

ITEM	PART / SPEC	WHY YOU NEED IT	QTY
Ariva-S-Mini scale (stand-alone)	Model 1070 weight-only ok	The scale itself. Weight-only version is enough for a pack-station weight check.	1
12 V wall-plug transformer	In 100–240 V Out 12 V / 0.84 A	Powers the Ariva-S. Usually ships in the box — confirm it is included, as the USB link does <u>not</u> power the scale.	1
Interface cable — USB RJ50 → USB-A	#72256236	Recommended for the tablet. USB virtual-COM cable (FTDI). Plugs the scale into the tablet's USB world.	1
Interface cable — RS-232 optional RJ50 → DB9	#72256235	Only if you connect to a PC COM port instead of the tablet. Needs a separate RS-232→USB adapter for Android.	0–1
USB-C OTG hub with PD passthrough	USB-C ↔ USB-A + USB-C PD in	The Tab A9 has one USB-C port. A PD hub runs the scale and keeps the tablet charged at an always-on station.	1
USB-C charger + cable	≥ 15 W PD	Feeds the hub's PD input so the mounted tablet never runs flat.	1
Mount / cable ties optional	—	Tidy the station and strain-relieve the cables.	—

Note A simple USB-C-to-USB-A OTG adapter also works if you don't need to charge while weighing — but for a permanently mounted station, use the PD hub.

CONNECTION DIAGRAM

How it all wires together



Data path is a single serial link: the scale streams weight frames up the USB cable; Android's USB-host stack presents it to the app as a serial device. No PC and no cloud sit in the middle.

CONFIGURATION — BOTH SIDES MUST MATCH

STEP 1 · ON THE SCALE

Scale soft-switch setup (Group 3)

Enter setup: hold the **>T<** (Tare) button ~5 s until CONF shows. **Zero** = confirm / next, **Tare** = step through groups. Open **Group 3** and set:

STEP	SETTING	CHOOSE
3.1	Communication Type	1 – USB Ser/CDC (Virtual COM) (use 0 – RS-232 only for the DB9/FTDI route)
3.2	Baud rate	3 – 9600
3.3	Parity	0 – None
3.4	Data bits	1 – 8 data bit
3.5	Stop bits	1 – 1 stop bit
3.6	Protocol Selection	3 – 8217 Mettler-Toledo (W0)

Step out to EXIT → SAVE to store and reboot. The scale is ready when the weight display returns.

App settings (Settings → Weight source)

- Set **Weight source** to **Serial — Streaming** (8217 WO streams continuously). If no reading appears, switch to **Serial — Polling**, which requests each reading with `W<␣`.
- Under **Serial connection**, match the scale exactly: 9600 / None / 8 / 1, protocol name 8217 Mettler-Toledo.
- Plug in the scale — Android shows a one-time "**Allow access to USB device?**" prompt. Tick *always* and accept.
- The reading flows into the shared **ON SCALE** value; every order compares live, exactly as in test mode.

Auto-connect The app opens the port on launch and also when the scale is plugged in later — no manual "connect" step.

Cable wiring (for reference / re-crimping)

You should not need to wire these yourself — buy the assembled Mettler-Toledo cables. Included for field diagnosis.

RS-232 · RJ50 → DB9 female #72256235

SCALE RJ50	DB9 FEMALE	SIGNAL
Pin 2	Pin 2	RxD — Receive Data
Pin 3	Pin 3	TxD — Transmit Data
Pin 4	Pin 5	GND — Signal Ground
Pin 7	Pin 8	CTS — Clear to Send
Pin 8	Pin 7	RTS — Request to Send

RJ50 pins 1, 5, 6, 9, 10 are proprietary (diagnostics, display power, loops) and are omitted for host integration.

USB · RJ50 → USB-A #72256236

SCALE RJ50	USB-A	SIGNAL
Pin 4	Pin 4	GND
Pin 5	Pin 3	D+ (Data+)
Pin 6	Pin 2	D- (Data-)
Pin 10	Pin 1	VCC — +5 V power detect

An FTDI bridge is molded into the cable shell, so the tablet sees a standard USB serial device.

Protocol, units & the driver page

- **Protocol 8217 Mettler-Toledo (WO).** "Weights-Only": the scale reports the live weight; the host can also send zero/tare commands. The app reads the weight frames and updates the shared reading.
- **Units.** The Ariva is metric and reports kilograms (e.g. 0.250). The app converts to grams automatically (a decimal value $\Rightarrow$ kg $\Rightarrow$ $\times 1000$; a bare integer $\Rightarrow$ grams; kg/g/lb tokens are also honoured).
- **The Ariva "Drivers" download page (JPOS / OPOS / Virtual COM Ports)** is for PC POS systems (Windows/Linux). You do **not** need any of those for the Android tablet — the app talks to the FTDI/CDC device directly. Only install the Virtual-COM (FTDI) driver if you later connect the scale to a Windows PC.

IF SOMETHING'S OFF

Troubleshooting

SYMPTOM	FIX
No reading / "connect a scale"	Confirm the scale is on (wall adapter), the OTG hub works, and you accepted the USB permission prompt. Re-check that the scale's Comm Type matches the cable (USB-CDC for #72256236). Try Serial — Polling if streaming shows nothing.
Weight is ~1000× off	Unit mismatch. The app assumes kg for decimal values; if your scale is set to output integer grams it's already handled. If still wrong, tell us the raw frame (below).
Garbled / no frames	Baud/parity/data/stop bits must match on both sides. Start with 9600 / None / 8 / 1; if garbled, try the classic MT set 9600 / Even / 7 / 1 on both.
Tablet won't charge	Use the USB-C PD hub and feed its PD-C input from the charger; a plain OTG adapter can't charge and host at once.
Reading raw frames	The driver logs each raw frame via <code>adb logcat</code> — useful for tuning if a non-standard framing appears.

Safety Always unplug power before connecting/disconnecting cabling. Use only genuine Mettler-Toledo cables — counterfeit assemblies can damage the scale and void warranty.

BRING-UP CHECKLIST

☐	Scale on mains via its 12 V adapter; weight display shows.
☐	Scale Group 3 set: USB-CDC · 9600 · None · 8 · 1 · Protocol 8217; SAVED.
☐	#72256236 cable → USB-C PD hub → tablet; charger into hub.
☐	App: Weight source = Serial (Streaming); serial values matched.
☐	USB permission accepted; a live weight appears in ON SCALE .
☐	Place a known weight → reading matches within tolerance.